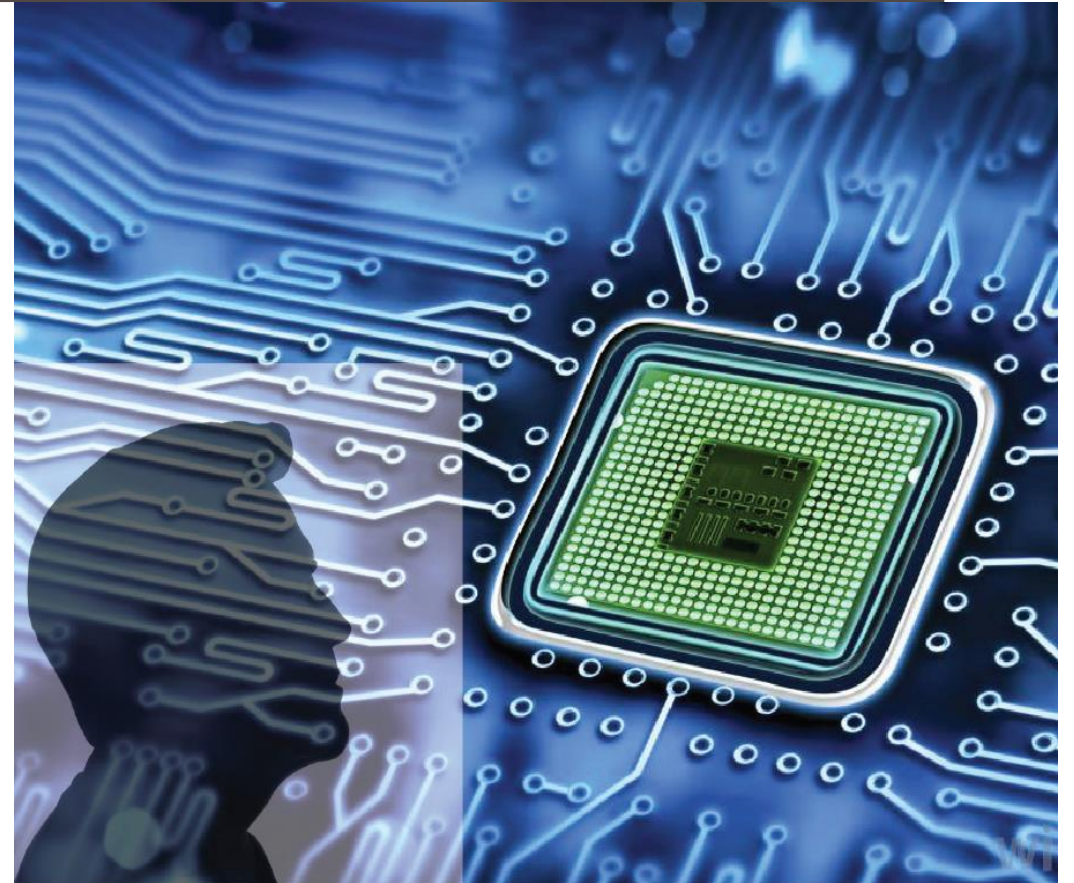


VLSI DESIGN FLOW

Week-9 Lecture-2 STA

Sneh Saurabh
16th October, 2024



VDF: Revision/Announcements

-



Static Timing Analysis

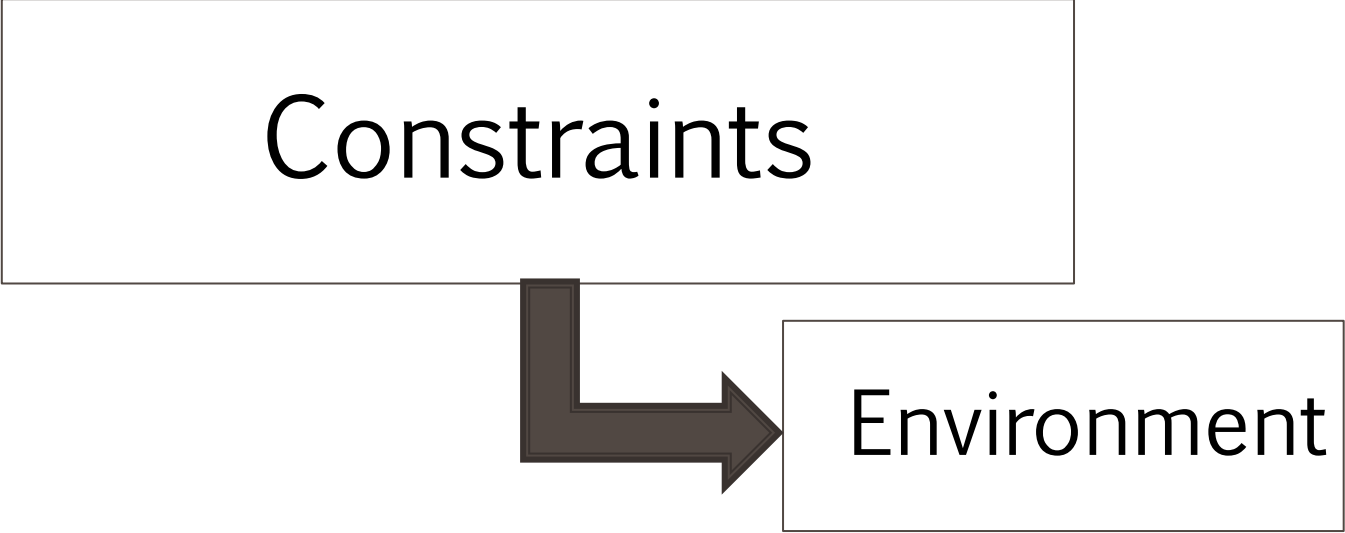


Constraints

Attributes of Clock Signal: Transition

set_clock_transition: specify an estimated clock transition time

```
create_clock -name CLK -period 2000 [get_ports clk_port]  
set_clock_transition 10 [get_clocks CLK]
```

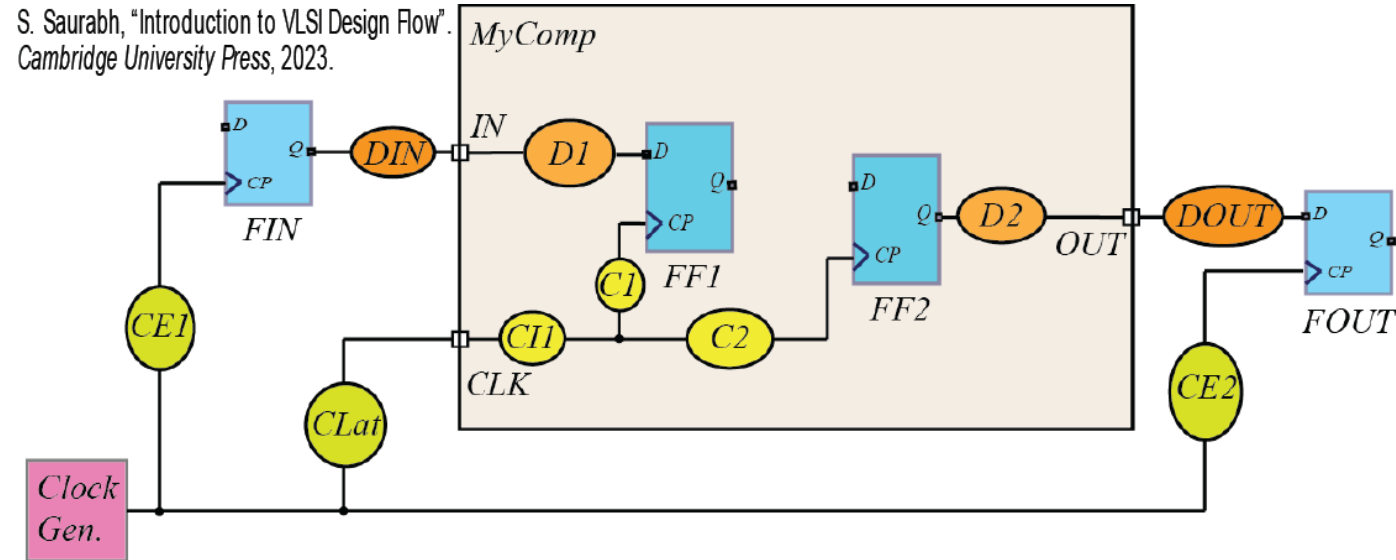


```
graph TD; A[Constraints] --> B[Environment]
```

Constraints

Environment

Static Timing Analysis: Environment of design

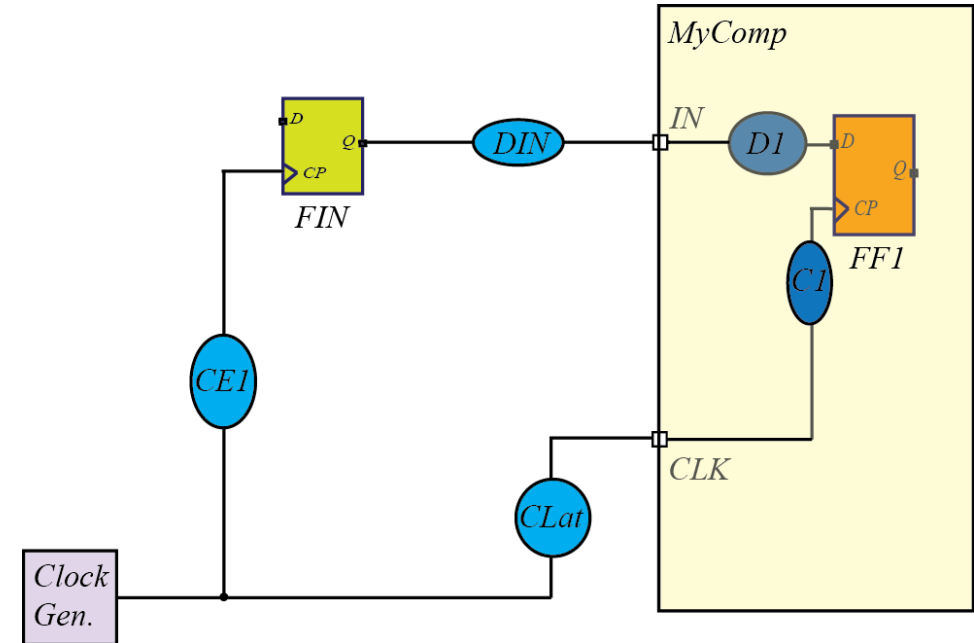


Environment of design puts additional constraints on the signal entering/leaving a design

- Signal entering the design at the input port encounters delay
- Signal leaving the design at the output port will encounter delay and get captured externally
 - Should meet setup/hold requirements at *FOUT*
 - Check for setup/hold must also be performed at the output port

Input Port Constraints: *set_input_delay*

- Signal gets delayed before entering the design
 - Lesser part of clock-period is available for the signal to reach flip-flop within the design
- Delay external to design at the input ports is modelled using *set_input_delay*



- Value of input delay = Delay of *CE1* + CP → Q delay of *FIN* + Delay of *DIN*

```
create_clock -name CLK -period 2000 [get_ports CLK_PORT]
```

```
set_input_delay -clock [get_clocks CLK] 100 [get_ports /M]
```

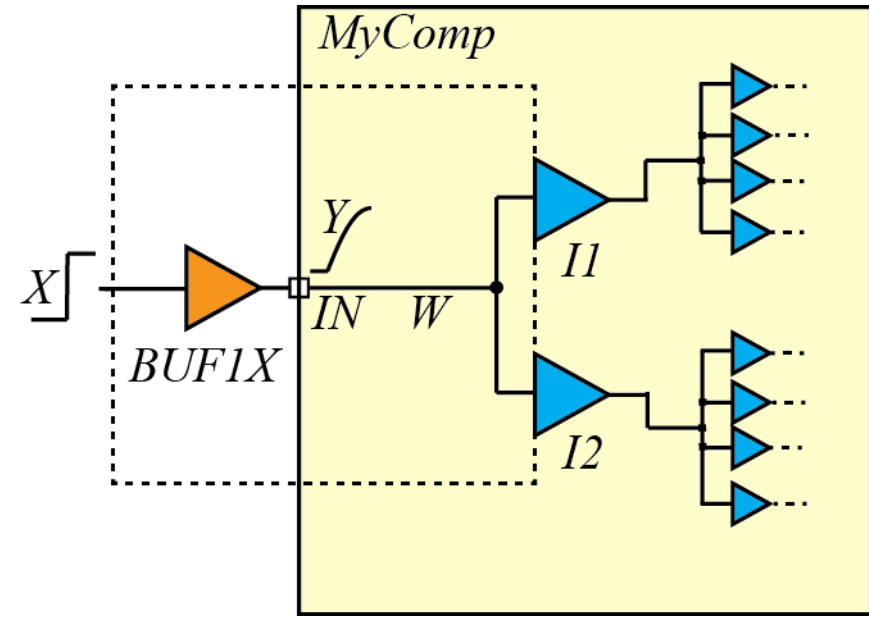
Transition at Input Port

- *set_input_transition*: model slew of incoming signal

```
set_input_transition 10 -max -rise [get_ports /M]
```

- *set_driving_cell*: driver of inputs

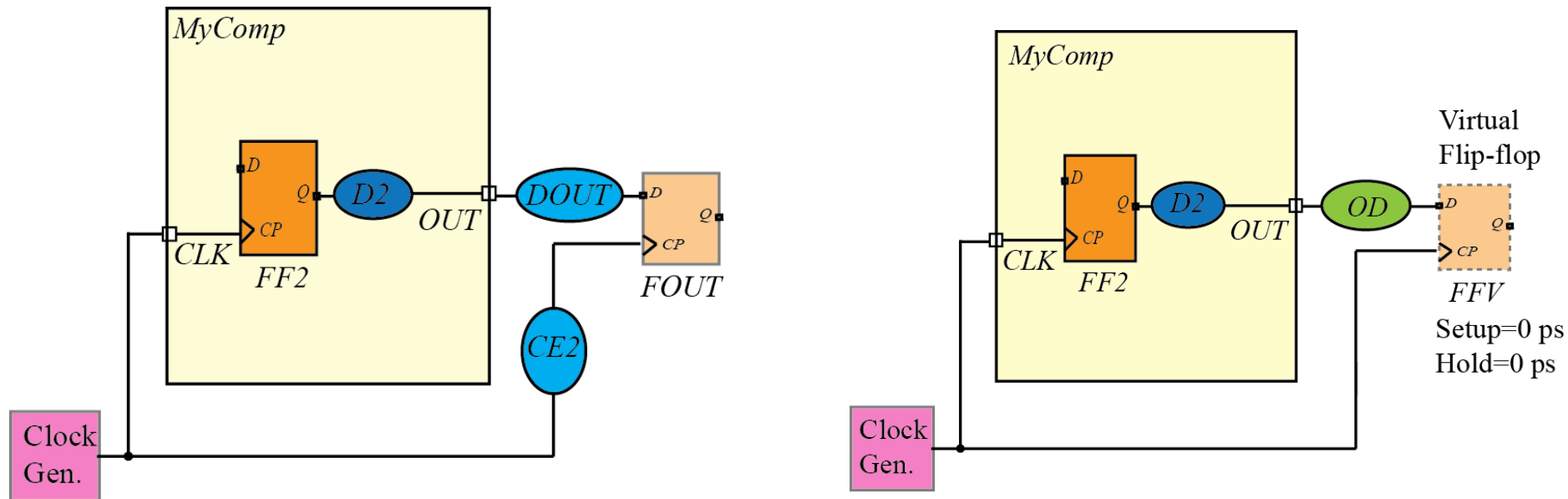
```
set_driving_cell -lib_cell BUF1X -library tech14nm  
[get_ports /M]
```



Output Port Constraints: *set_output_delay*

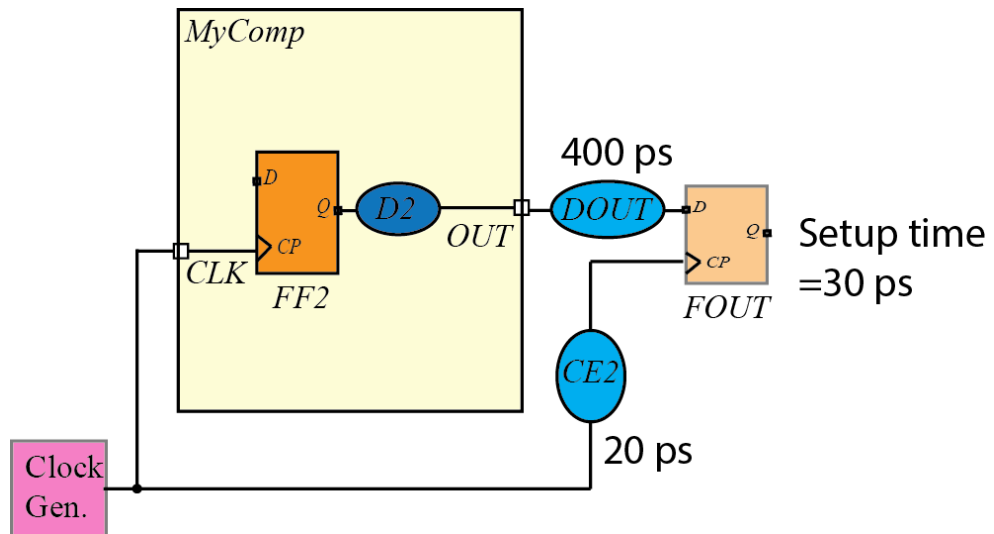
Output Delay: signal leaving a design must meet the setup/hold constraints of the flip-flop that captures that signal

- Constraints of the external flip-flop is modelled using *set_output_delay*



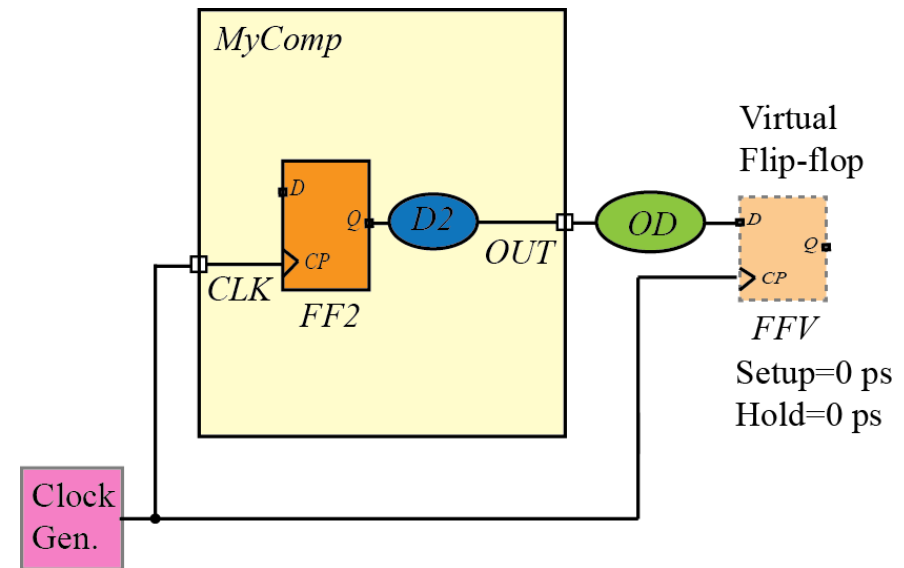
- The delay of OD needs to be chosen such that the setup/hold requirements in the actual circuit and equivalent circuit match
- Delay of OD is specified in *set_output_delay* command

Illustration: *set_output_delay*



$$T_{int} + 400 < T_{clk} + 20 - 30$$

$$T_{OD} = 400 - 20 + 30 = 410$$



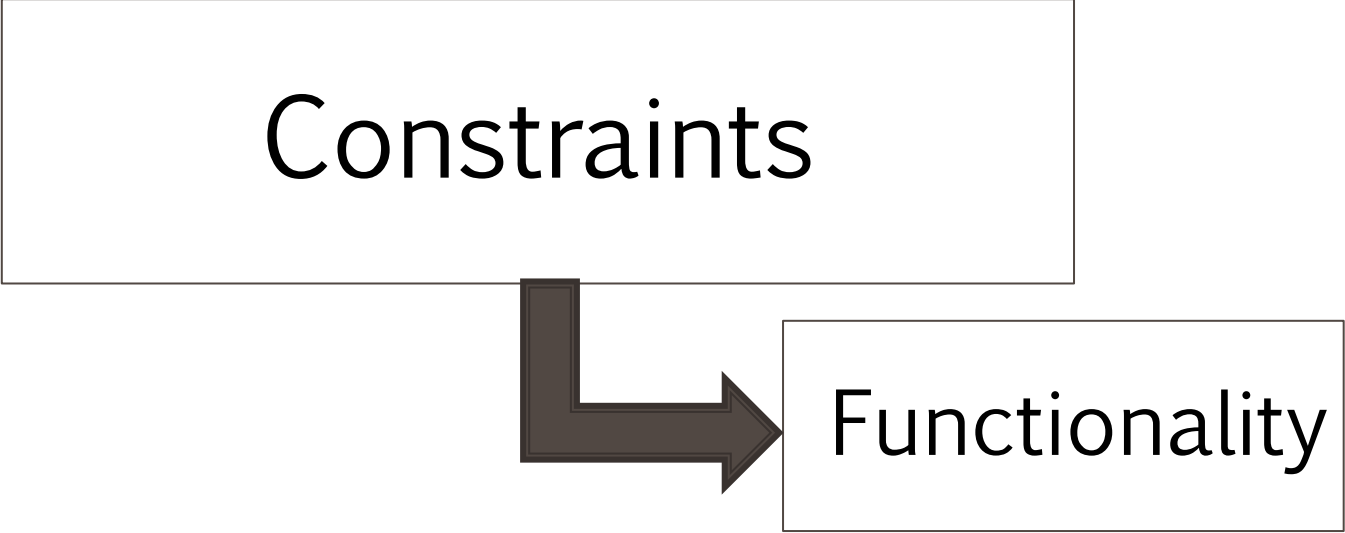
$$T_{int} + T_{OD} < T_{clk}$$

```
create_clock -name SYS_CLOCK -period 2000 [get_ports CLK]
set_output_delay 410 -max -clock [get_clocks SYS_CLK] [get_ports OUT]
```

Load at Output Port

- *set_load*: load that will be driven by the output port

```
set_load 0.039 [get_ports OUT]
```



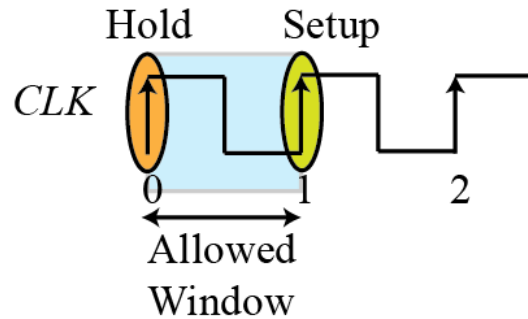
```
graph TD; A[Constraints] --> B[Functionality]
```

Constraints

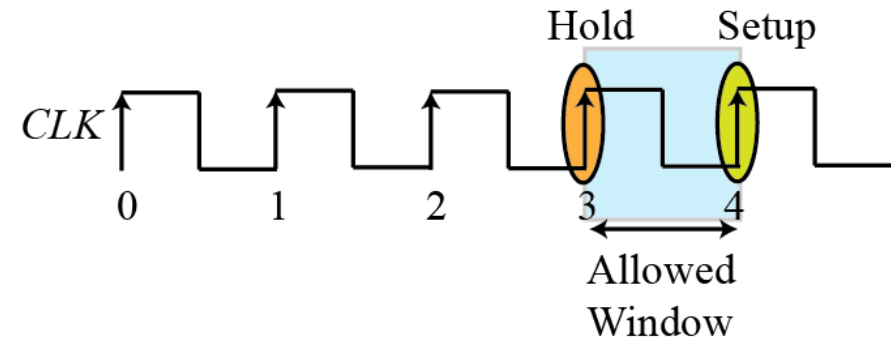
Functionality

Timing Exceptions

- *set_false_path*: To make exceptions from analysing certain paths that may not be exercised
- *set_multicycle_path*: To inform the STA tool that certain path may take more than one cycle

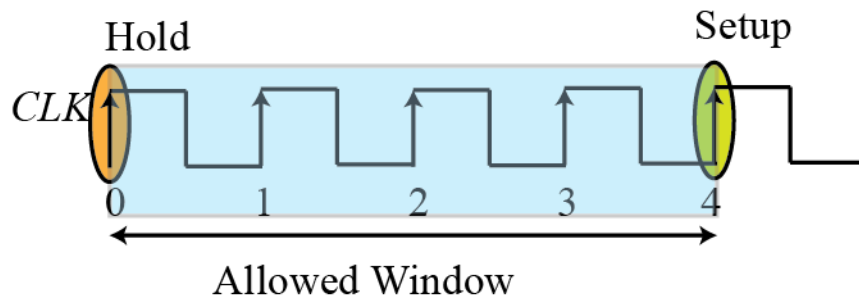


```
set_multicycle_path 4 -setup -from [get_pins FF1/CP]  
-to [get_pins FF2/D]
```



```
set_multicycle_path 4 -setup -from [get_pins FF1/CP]  
-to [get_pins FF2/D]
```

```
set_multicycle_path 3 -hold -from [get_pins FF1/CP]  
-to [get_pins FF2/D]
```



Constant Value to Port/Pin

- *set_case_analysis*: assign constant value to some port/pin

```
set_case_analysis 1 [get_ports SCAN_ENABLE]
```

```
set_case_analysis 1 [get_ports SLEEP_MODE]
```

References

- S. Saurabh, “Introduction to VLSI Design Flow”. Cambridge: *Cambridge University Press*, 2023.
- Bhasker, Jayaram, and Rakesh Chadha. *Static timing analysis for nanometer designs: A practical approach*. Springer Science & Business Media, 2009.